TESLA COACHING CLASSES

Mathematics – Part 2 (Geometry)

Time: 2 Hours	Standard: Class 10	Total Marks: 40

Part Syllabus Test - 06 (PST-06)

Student Name: _____ **Date:** _____

CHAPTERS COVERED & WEIGHTAGE

CHAPTER 4	Geometric Constructions	60%
CHAPTER 3	Circle	30%
CHAPTER 1	Similarity	5%
CHAPTER 2	Pythagoras Theorem	5%

Q.1(A) **Choose the correct alternative from given:** **[4 Marks]**

i) The number of tangents that can be drawn to a circle at a point on the circle is _____.

(a) 3 **(b)** 2

(c) 1 **(d)** 0

ii) If $\Delta ABC \sim \Delta PQR$ and $\frac{AB}{PQ} = \frac{7}{5}$, then _____.

(a) ΔABC is bigger. **(b)** ΔPQR is bigger.

(c) Both triangles will be equal. **(d)** Can not be decided.

iii) Which theorem is used while constructing a tangent to the circle by using center of a circle?

(a) Tangent – radius theorem. **(b)** Converse of tangent – radius theorem.

(c) Pythagoras theorem. **(d)** Converse of Pythagoras theorem.

iv) Two circles of radii 5.5 cm and 3.3 cm respectively touch each other. What is the distance between their centers?

(a) 4.4 cm **(b)** 8.8 cm

(c) 2.2 cm **(d)** 8.8 or 2.2 cm

Q.1(B) **Solve the following subquestions:** **[4 Marks]**

i) Construct $\angle PQR = 115^\circ$ and divide it into two equal parts.

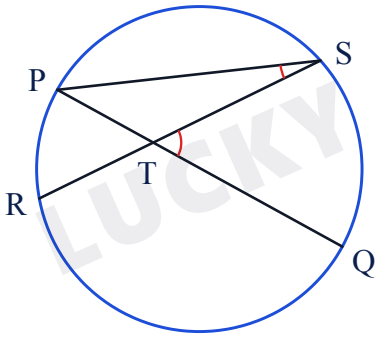
ii) Draw seg AB of length 4.5 cm and draw its perpendicular bisector.

iii) Draw seg AB of length 9 cm and divide it in the ratio $3 : 2$.

iv) How many circles can be drawn passing through a point?

Q.2(A)	Complete and write any TWO activities from the following:	[4 Marks]

i) In the figure, if the chord PQ and chord RS intersect at point T. Prove that $m\angle \mathrm{STQ} = \frac{1}{2}[m(\text{arc PR}) + m(\text{arc SQ})]$ for any measure of $\angle \mathrm{STQ}$ by filling out the boxes.

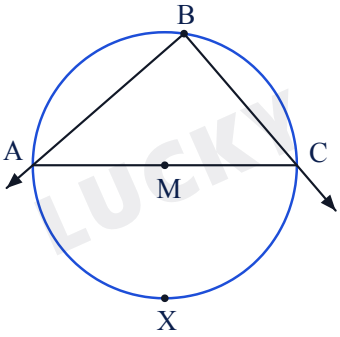


Proof: $m\angle \mathrm{STQ} = m\angle \mathrm{SPQ} +$ ☐ ---[Theorem of the external angle of a triangle]

$= \frac{1}{2} m(\text{arc SQ}) +$ ☐ ---[Inscribed angle theorem]

$= \frac{1}{2}$ [☐ + ☐]

ii) The angle inscribed in the semicircle is a right angle. Prove the result by completing the following activity.



Given: $\angle \mathrm{ABC}$ is inscribed angle in a semicircle with center M .

To prove: $\angle \mathrm{ABC}$ is a right angle.

Proof: Segment AC is a diameter of the circle.

$\therefore m(\text{arc AXC}) =$ ☐ ---(i) [Measure of semi circular arc is 180°]

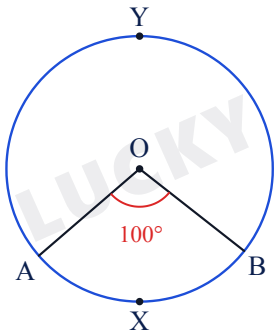
Arc AXC is intercepted by the inscribed angle $\angle \mathrm{ABC}$

$\angle \mathrm{ABC} =$ ☐ ---[Inscribed angle theorem]

$= \frac{1}{2} \times$ ☐ $=$ ☐

$\therefore \angle \mathrm{ABC}$ is a right angle.

iii) If O is the center of the circle in the figure alongside, then complete the table from the given information.

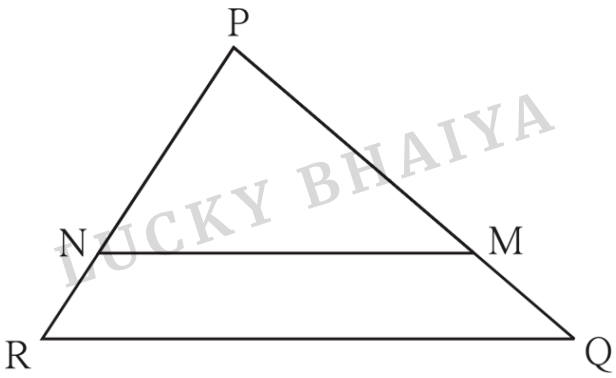


Type of arc	Name of the arc	Measure of the arc
Minor arc		
Major arc		

Q.2(B) **Solve any FOUR subquestions from the following:** **[8 Marks]**

i) Construct a tangent to a circle with centre P and radius 3.2 cm at any point M on it.

ii) In ΔPQR, $PM = 15$, $PQ = 25$, $PR = 20$, $NR = 8$. State whether line NM is parallel to side RQ. Give reason.



iii) Draw a circle of radius 3.6 cm. Draw a tangent to the circle at any point on it without using the centre.

iv) Find the length of the side and perimeter of an equilateral triangle whose height is $\sqrt{3}$ cm.

v) Draw a circle with a diameter AB of length 6 cm. Draw a tangent to the circle from the endpoints of the diameter.

Q.3(A)	Complete and write any ONE activity from the following:	[3 Marks]

i) $\square MRPN$ is cyclic, $\angle R = (5x - 13)^\circ$, $\angle N = (4x + 4)^\circ$. Find measures of $\angle R$ and $\angle N$, by completing the following activity.

$\square MRPN$ is a cyclic quadrilateral. ---[Given]

Opposite angles of a cyclic quadrilateral are [ ] .

$\therefore \angle R + \angle N =$ [ ]

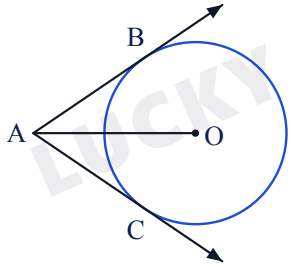
$\therefore (5x - 13)^\circ + (4x + 4)^\circ =$ [ ]

$\therefore 9x - 9 = 180 \implies 9x = 189 \implies x =$ [ ]

$\therefore \angle R = (5x - 13)^\circ = (5 \times 21 - 13)^\circ =$ [ ]

$\therefore \angle N = (4x + 4)^\circ = (4 \times 21 + 4)^\circ =$ [ ]

ii) In the given figure, O is the centre of the circle. Seg AB, seg AC are tangent segments. Radius of the circle is r and $l(AB) = r$, Prove that $\square ABOC$ is a square.



$l(AB) = r$ ---[Given] ---(i)

$AB = AC$ --- [ ] ---(ii)

But $OB = OC = r$ --- [ ] ---(iii)

From (i), (ii) and (iii), $AB =$ [ ] $= OB = OC = r$

$\therefore$ Quadrilateral $ABOC$ is [ ]

Similarly, $\angle OBA =$ [ ] ---[Tangent theorem]

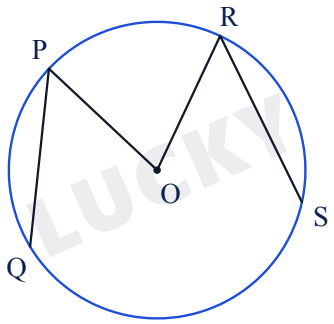
If one angle of [ ] is right angle, then it is a square.

$\therefore$ Quadrilateral $ABOC$ is a square.

Q.3(B)	Solve any TWO subquestions from the following:	[6 Marks]

i) $\Delta ABC \sim \Delta LMN$. In ΔABC, $AB = 5.5$ cm, $BC = 6$ cm, $CA = 4.5$ cm. Construct ΔABC and ΔLMN such that $\frac{BC}{MN} = \frac{5}{4}$.

ii) In figure, O is the centre of a circle, chord PQ $\cong$ chord RS. If $\angle POR = 70^\circ$ and $(\text{arc RS}) = 80^\circ$, find (i) $m(\text{arc PR})$ (ii) $m(\text{arc QS})$ (iii) $m(\text{arc QSR})$



iii) Draw a circle with centre P and radius 3.4 cm. Take point Q at a distance 5.5 cm from the centre. Construct tangents to the circle from point Q.

iv) Draw a circle of radius 4.2 cm. Draw arc PQ measuring 120°. Draw a tangent to the circle from point P and point Q.

Q.4	**Solve any TWO subquestions from the following:**	**[8 Marks]**

i) $\Delta ABC \sim \Delta LBN$. In ΔABC, $AB = 5.1$ cm, $\angle B = 40^\circ$, $BC = 4.8$ cm, $\frac{AC}{LN} = \frac{4}{7}$. Construct ΔABC and ΔLBN.

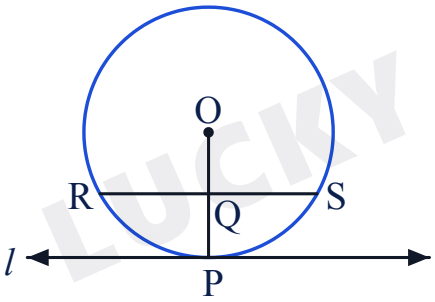
ii) Draw a circle with center O and radius 2.8 cm. Take point P in the exterior of a circle such that tangents PA and PB drawn from point P make an angle $\angle APB$ of measure 70°.

iii) Construct ΔPYQ such that, $PY = 6.3$ cm, $YQ = 7.2$ cm, $PQ = 5.8$ cm. If $\frac{YZ}{YQ} = \frac{6}{5}$, then construct ΔXYZ similar to ΔPYQ.

Q.5	**Solve any ONE subquestion from the following:**	**[3 Marks]**

i) Point P is at a distance of 6 cm from line AB. Draw a circle of radius 4 cm passing through point P so that line AB is the tangent to the circle.

ii) In the figure, line l touches the circle with center O at point P. Q is the midpoint of radius OP. RS is a chord through Q such that chords RS $\parallel$ line l. If $RS = 12$, find the radius of the circle.



LUCKY

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